

Obesity Model Prediction

School of Stats

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1 Introduction

Obesity has become an increasingly dangerous trend worldwide. In 2022, one in eight adults (890 million people) were living with obesity, and 43% of adults were classified as overweight. Since 1990, the global prevalence of adult obesity has more than doubled, while adolescent obesity has quadrupled. This surge raises healthcare costs and fuels conditions such as type 2 diabetes, cardiovascular disease, and certain types of cancer. Early risk identification through automated tools could enable targeted lifestyle interventions, reducing long-term disease burden and improving quality of life.

The "Obesity Risk Prediction" competition on Kaggle offers demographic, dietary, and behavioral data from more than 4,000 individuals, providing a real-world testbed for machine learning. In this project, I develop and validate a predictive pipeline that classifies each participant into one of the seven levels of obesity using categorical and numeric features. The goal is to figure out if I could accurately predict an individual's obesity category (ranging from underweight to obesity type III) based off of patterns in their lifestyles and biological factors.

1.1 Approach Overview

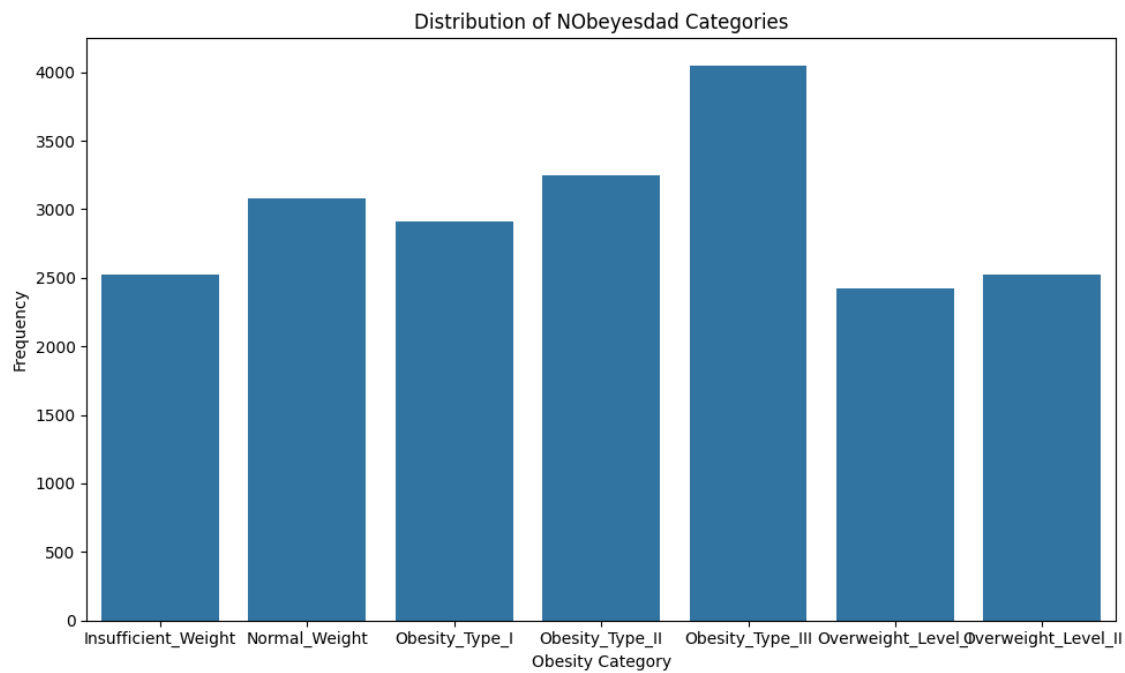
1. Data preparation: Compute BMI from height and weight, encode categorical features, and split dataset into training and validation sets
2. Modeling: Train four classifiers: logistic regression, decision tree, random forest, and LightGBM. Assessed via 5-fold stratified cross-validation
3. Evaluation: Evaluate each model to determine the strongest and examine the confusion matrix to understand misclassifications patterns

2 Data Overview

The core dataset for this project was generated by a deep-learning model that had been pre-trained on a larger, survey-based dataset. The original survey was deployed via a web platform, where anonymous respondents from Mexico, Peru, and Colombia (ages 14-61) answered questions about their demographics, lifestyle, and physical condition. The survey responses were processed through the deep-learning pipeline to yield 17 standardized attributes and an estimated obesity category for each individual. The competition provided both the training set (20758 entries) and the test set (13840 entries).

Each record contains seven numeric variables: age, height, weight, frequency of vegetable consumption (FCVC), number of main meals (NCP), daily water intake (CH2O), frequency of physical activity (FAF), and hours of screen time (TUE). There are also nine categorical variables that cover gender, family history of overweight, high-calorie food consumption (FAVC), between-meal eating (CAEC), smoking status, calorie counting (SCC), alcohol consumption (CALC), and primary transportation mode (MTRANS). I add a computed Body Mass Index ($BMI = \text{weight} / \text{height}^2$) and an ordered target variable (NObeyesdad) with seven levels from Insufficient Weight through Obesity Type III.

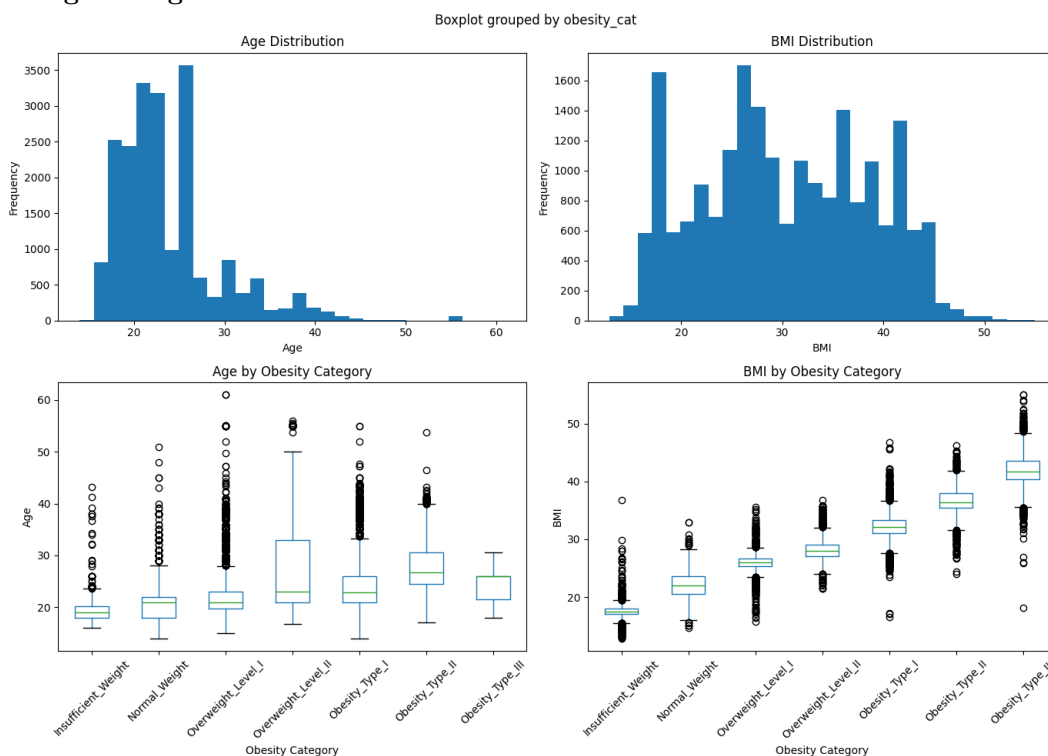
Because every survey response was processed by the same pre-trained model, the resulting dataset arrived fully labeled. This meant that there weren't any null values or duplicates that I needed to fix, so there wasn't much to the data cleaning process. I chose to encode the categorical variables and compute BMI because the model requires all numeric inputs, and BMI is a well-established composite measurement that captures the relationship between height and weight in a single feature.



Looking at the distribution of the response variable (NOBeyesdad), I would say that it is pretty balanced. Only Obesity Type III showed above-average frequencies. To ensure that each of these seven classes appears in every training and validation fold, I used StratifiedKfold. I did so to preserve the overall class proportions in each split, preventing any fold from under- or over-representing certain obesity levels and ensuring that the cross-validation metrics accurately reflect performance across all categories.

2.1 EDA

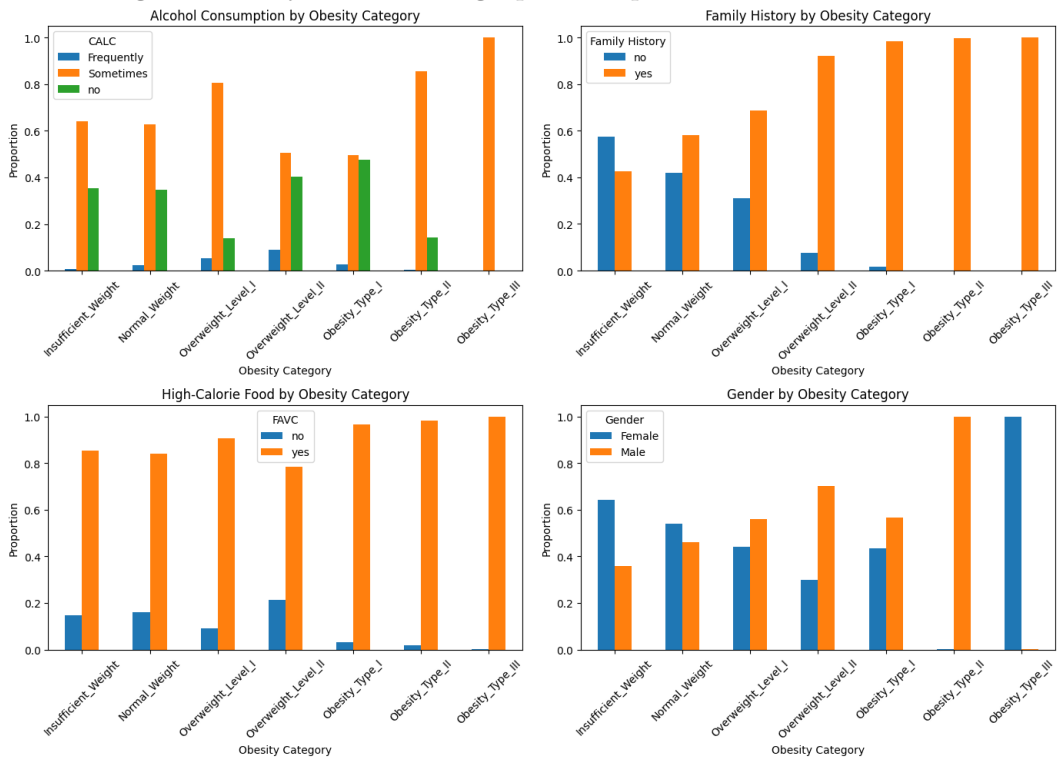
First Figure: Age and BMI EDA



1. Age histogram shows a roughly bell shaped with a right skew. This means that there is a higher frequency of younger respondents than older.
2. BMI histogram spans from underweight ($< 18.5 \text{ kg/m}^2$) up to extreme obesity ($> 45 \text{ kg/m}^2$). Semi-balanced with some outliers.

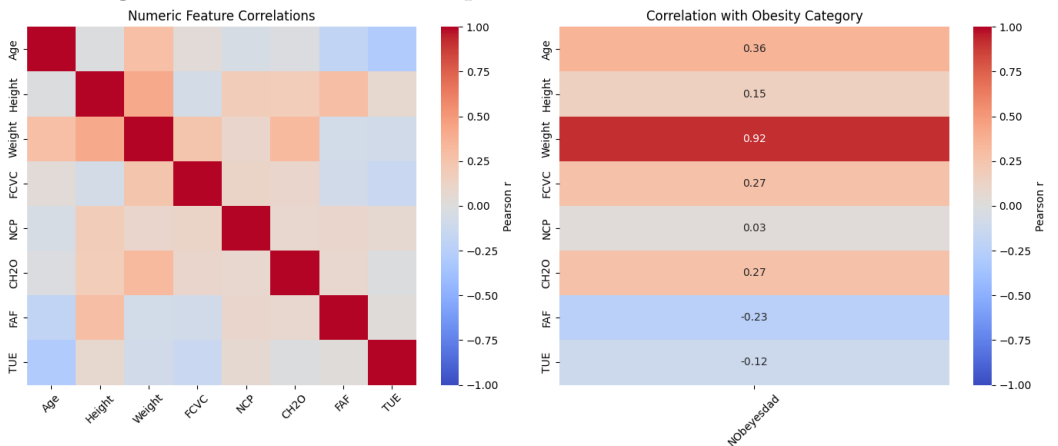
3. Outliers in age (above 60) and BMI are still realistic. I did not remove or limit them because the models can handle a few extreme values, and keeping them helps the model learn from full range of cases.

Second Figure: Lifestyle and Demographic Proportions



1. Alcohol Consumption (CALC) shifts from a mix of "no" and "sometimes" in lower-risk groups to almost exclusively "sometimes" among the highest obesity levels.
2. Family History of Overweight jumps from around 40% "yes" in normal weight to over 95% "yes" across all obesity categories, indicating strong familial patterns of obesity in overweight people.
3. High-Calorie Food Intake (FAVC) rises from roughly 80% "yes" in the lower categories to above 95% in Obesity Type II and III, highlighting diet's role in obesity control.
4. Gender Proportions flip from a female majority at the extremes (Insufficient Weight and Obesity Type III) to a male majority in the mid-range obesity levels. This suggests that there are potential sex-specific risk factors.

Third Figure: Correlation Heatmap



1. The left correlation heatmap is a pairwise Pearson correlation among the eight numeric features. It reveals that Weight and Height have a higher correlation among other features. This led to the creation of the BMI feature. The BMI feature is calculated by taking the Weight (kg) divided by Height². Most other

values have a relatively low correlation. This gives me confidence that each numeric feature contributes largely unique information.

2. The right heatmap shows each feature's correlation with the ordered obesity category "NObesidad". Weight stands out as the strongest linear predictor of obesity level. Other features such as Age, FCVC, and CH20 have moderate positive relationships, suggesting older age and higher ewater/vegetable intake also trend with higher obesity categories.

3 Methodology

3.1 Data Splitting

To assess generalization, the training data was randomly split into 80% for model fitting and 20% for validation, using stratified sampling to preserve the seven-level obesity distribution for both sets. In addition, all cross-validation experiments employed 5-fold StratifiedKFold, ensuring each fold maintains the same class proportions as the full dataset.

3.2 Pre-processing

All categorical features are one-hot encoded so that each level becomes a binary indicator. Numerical features, including the engineered feature BMI, are standardized to a zero mean and unit variance. This uniform scaling benefits models that assume centered data (e.g. logistic regression) and speeds up gradient-based training.

3.3 Model Choices

1. Multinomial Logistic Regression

I include multinomial logistic regression as a transparent baseline because by modeling the log-odds of each obesity level as a linear function of the predictors, its coefficient estimates directly reveal how a one-unit change in features (such as BMI or age) shifts the odds of moving between categories. This interpretability helps diagnose data issues and establishes a performance floor against which to compare to more complex models.

2. Decision Tree

The Decision Tree captures simple non-linear interactions while remaining easy to visualize and explain. I limit the tree depth to avoid over-splitting noise and to keep decision rules concise.

3. Random Forest

Random Forest builds on trees by averaging many decorrelated models, smoothing out errors, and usually boosting accuracy. I tune its number of trees and depth to find the best bias-variance trade-off without losing the ability to extract feature-important insights.

4. LightGBM

LightGBM uses gradient boosting with histogram binning and leaf-wise growth to achieve high performance on tabular data. By borrowing optimal hyperparameters from a Kaggle discussion, I leverage well-tested settings and avoid an expensive and long search, saving time while still capturing complex interactions

3.4 Evaluation Metrics

1. Accuracy

Calculates the overall percentage of correct predictions across all seven obesity categories. Because the classes are overall pretty balanced, accuracy gives a clear, single-number summary of model performance.

2. Macro-F1

Computes precision and recall for each obesity level, then averages their F1 scores without weighting by class size. This ensures that performance on both common and less common categories is treated equally, preventing high accuracy on the largest class from masking poor performance elsewhere.

3. **AUC-ROC**

Treats each obesity level in turn as the positive class against all others and measures the area under the ROC curve. Averaging these seven AUCs evaluates how well the model ranks each category independently of any specific decision threshold.

4. **Confusion Matrix**

The confusion matrix offers a detailed view of the model’s performance by showing how often each true obesity level is assigned to each predicted level. It reveals the pattern of misclassifications, such as whether certain categories are systematically confused with their neighbors.

4 **Result**

Below, I compare the four models’ performance on the held-out validation split and 5-fold stratified cross-validation, highlight key patterns in their errors and strengths, and explain the final model choice.

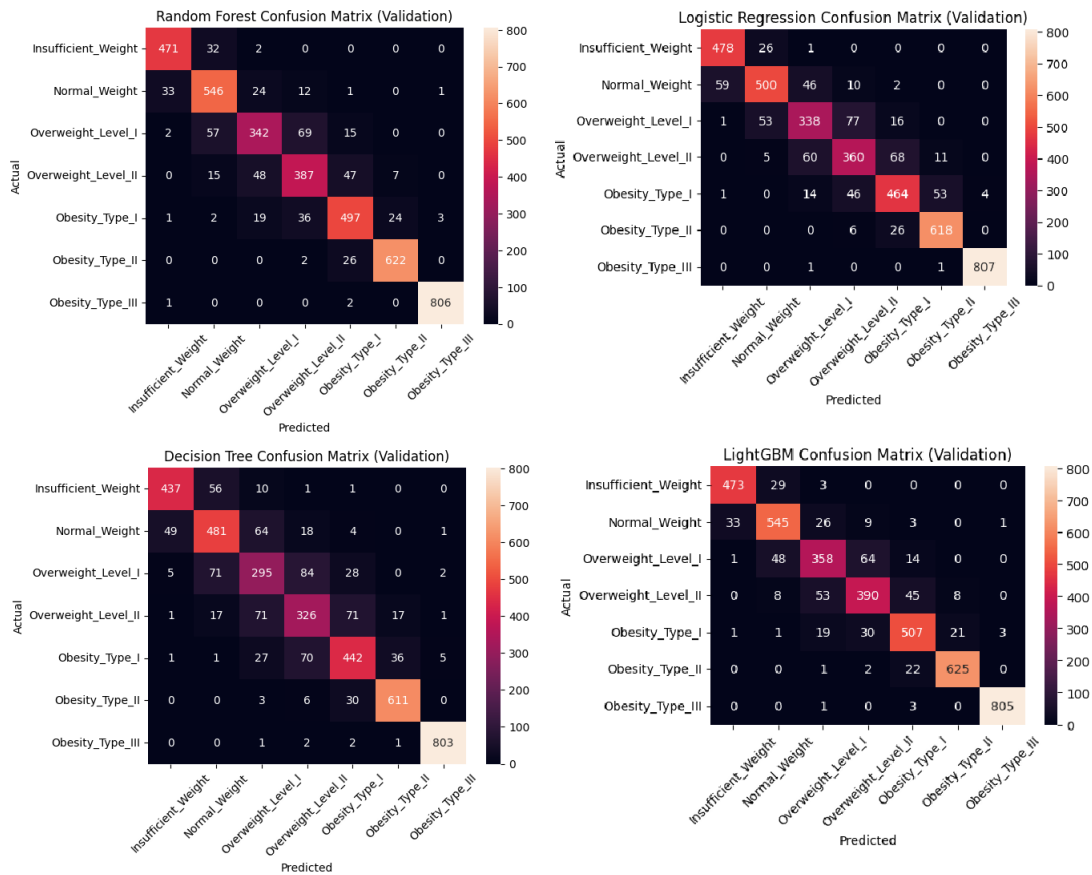
4.1 **Performance Metrics**

Model	5-Fold CV	Accuracy	Macro-F1	Weighted-F1	ROC-AUC
Logistic Regression	0.852 ± 0.006	0.859	0.84	0.86	0.978
Decision Tree	0.819 ± 0.005	0.818	0.80	0.82	1.000
Random Forest	0.876 ± 0.005	0.884	0.87	0.88	1.000
LightGBM	0.890 ± 0.004	0.892	0.88	0.89	0.998

Logistic regression establishes a clear baseline, capturing most of the linear separability in the data. The decision tree, while interpretable, underperforms on both accuracy and Macro-F1 and shows signs of overfitting (perfect ROC-AUC). Random forest improves accuracy and Macro-F1 substantially but also overfits the validation set. LightGBM strikes the best balance: it achieves the highest cross-validation and validation accuracy, a balanced Macro-F1 of 0.88, and a near-perfect ROC-AUC of 0.998 without fully memorizing the validation split.

I achieved a validation accuracy of 89.2 % with LightGBM, which means that for nearly nine out of ten people in my held-out set, the model correctly assigned their exact obesity category—ranging from Insufficient Weight through Obesity Type III—using only age, BMI, eating habits, physical activity, screen time, and family history. In practical terms, this high level of precision shows that these predictors carry very strong signals. In most cases, simply knowing someone’s BMI and related lifestyle factors is enough information for my algorithm to place them in the correct weight status group. For my scientific question—“Can I accurately predict an individual’s obesity category using these demographic and lifestyle features?”—this 89 % success rate demonstrates that the inputs and modeling approach I used can reliably identify obesity risk. Although about one in ten cases still falls near the threshold between two adjacent categories (for example, Overweight Level II versus Obesity Type I), this performance indicates that a LightGBM-based screening tool could form a dependable basis for early intervention or personalized health recommendations.

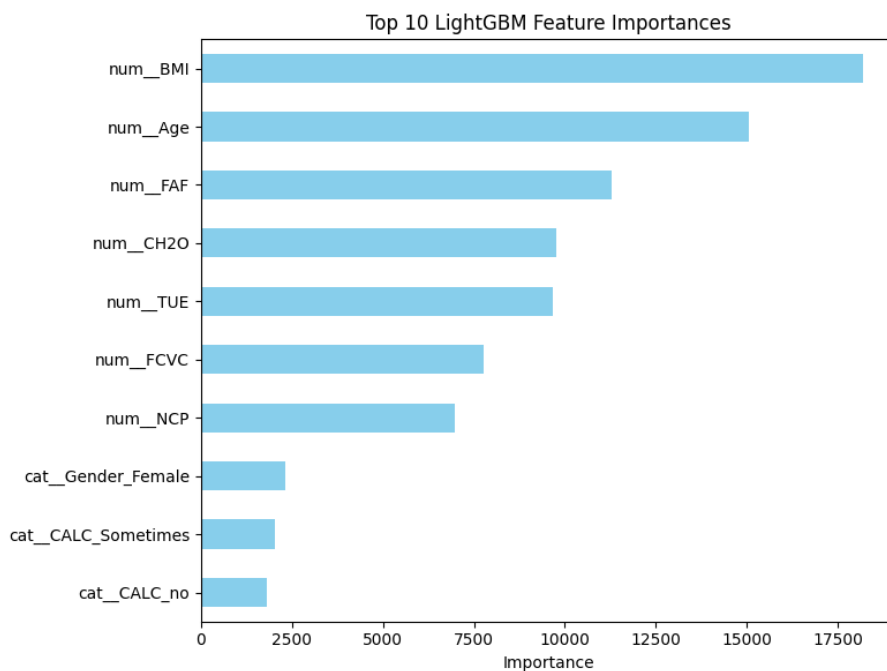
4.2 Confusion Matrix Insights



All models most often confuse adjacent categories with one another, reflecting the continuous nature of BMI-based thresholds. There were also the fewest errors at predicting Obesity Type III, indicating the model reliably identifies severe-obesity cases. Among them, LightGBM also shows the fewest off-diagonal entries, confirming its superior ability to distinguish between neighboring obesity levels.

4.3 Important Features

Below is the bar chart showing the top ten features as ranked by LightGBM's own importance scores:



The bar chart above shows LightGBM’s ranking of the top ten predictors on a scale of relative influence. At the very top sits BMI, which alone captures the bulk of the model’s decision-making; this is expected, since BMI is itself derived from height and weight and directly reflects a person’s body fat. Next in line is age, indicating that older participants were more likely to fall into higher obesity categories. Closely following are aspects of daily habits: the frequency of physical activity (FAF) comes third, highlighting that those who exercise more frequently tend to have lower obesity risk, and daily water intake (CH2O) and hours of screen time (TUE) occupy the fourth and fifth spots, respectively. In both cases, higher water consumption is associated with lower risk, while more screen time correlates with higher risk. After these lifestyle measures, frequency of vegetable consumption (FCVC) and number of main meals (NCP) show moderate importance, confirming that diet composition and meal regularity still contribute distinctly to classifying weight categories. Finally, three categorical indicators appear: Gender Female, CALC Sometimes (occasional alcohol consumption), and CALC no (no alcohol consumption). The fact that the female gender has a larger influence than most other categorical flags suggests that gender differences should be accounted for when predicting obesity risk, and the presence or absence of alcohol intake also shifts the probability toward one obesity level versus another.

Altogether, these ten features align closely with what we would expect from existing medical and nutritional knowledge: body-mass index and age remain the strongest single factors, but lifestyle patterns around exercise, hydration, screen use, meal habits, and even alcohol use provide the refinement necessary to distinguish between adjacent obesity categories. In relation to my scientific question—can I accurately predict an individual’s obesity category from demographic and lifestyle data?—these importance scores confirm that I have chosen the right set of inputs. Because BMI and age provide a clear baseline, and the additional behaviors fill in the gaps where BMI alone might place someone near a threshold, the model’s reliance on these top ten predictors demonstrates exactly why LightGBM can achieve nearly 90 % accuracy: it learns the core relationships and then uses the finer daily-habit signals to resolve borderline cases.

Based on these comprehensive evaluations, I selected the tuned LightGBM model as the final algorithm. It combines top predictive performance, reasonable generalization, and actionable insights into feature relevance.

5 Discussion and Reflection

Reflecting on this work, several strengths stand out. LightGBM consistently achieved the best performance, reaching about 89 % accuracy, a Macro F1 of 0.88, and a very high ROC AUC of 0.998. This shows that age, BMI, eating habits, physical activity, screen time, and family history together carry very strong signals. Comparing univariate effect sizes (Cramér’s V and n^2) to LightGBM’s feature importances confirmed that BMI and age dominate, while daily habits such as exercise frequency, hydration, and screen time provide the fine distinctions needed to separate adjacent categories. Using 5-fold Stratified KFold ensured that each obesity level was represented in every split, and the close match between cross-validation and validation results suggests minimal overfitting.

Nevertheless, several limitations deserve attention. First, I never had access to clinically measured obesity labels because all categories were generated by a pretrained deep learning model on a smaller survey. If that initial pipeline misclassified particular age groups or lifestyle patterns, LightGBM simply learned those biases. Second, collapsing height and weight into a single BMI variable assumes that no extra information remains in the raw measurements. In practice, two people with identical BMIs but different statures can have very different body compositions, which the model cannot distinguish. Third, the entire dataset comes from respondents in Mexico, Peru, and Colombia, so it is unclear whether the same relationships hold in other countries or demographic groups. Finally, I did not check whether the predicted probabilities are well calibrated; in a real-world screening tool, a 70 % probability should correspond to a 70 % chance of severe obesity, and miscalibration could lead to poor intervention decisions.

Several methodological challenges also arose. Tuning LightGBM’s hyperparameters from scratch on seventeen predictors proved extremely time-intensive, so I relied on optimized values from a high-scoring Kaggle notebook rather than exploring every combination. This shortcut saved hours, but limited my understanding of how each parameter interacts with this specific data. Another recurring issue was borderline cases—individuals whose BMI and habits place them on the cusp between two adjacent categories. No matter how well the model performed overall, those edge instances remained the hardest to classify because a single label cannot fully capture their uncertainty. Despite these drawbacks, the analysis still offers a robust, interpretable pipeline for predicting obesity categories within similarly structured data.

6 Conclusion

In this project, I set out to determine whether demographic and lifestyle features could accurately predict a person's obesity category. By comparing multiple classifiers—logistic regression, decision tree, random forest, and LightGBM—I found that LightGBM consistently delivered the strongest results, achieving nearly 90 percent accuracy and a Macro F1 of 0.88 on held-out data. BMI and age emerged as the most influential predictors, with exercise frequency, hydration, and screen time providing the finer distinctions needed to accurately separate adjacent weight-status groups.

My original question—can I predict an individual's obesity category using age, BMI, eating habits, physical activity, screen time, and family history?—was answered affirmatively. The model's high performance suggests that these inputs contain sufficient information to classify weight status reliably. In practical terms, this pipeline could be used to flag individuals at risk for overweight or obesity, enabling early lifestyle interventions or targeted recommendations.

Looking forward, obtaining clinically measured labels would strengthen confidence in any real-world application, and calibrating predicted probabilities would support risk-based decision making. Exploring continuous risk scores instead of discrete categories could also help manage borderline cases more gracefully. Overall, I learned that a carefully chosen set of features combined with a robust boosting algorithm can form a dependable tool for obesity risk prediction and pave the way for more personalized public health strategies.

References

1. [Kaggle Discussion - LGBM Hyperparameter Tuning](#)
2. [Kaggle Competition - Multi-Class Prediction of Obesity Risk Data](#)